MTDF Validation Suite Appendix: Reproducibility Index and Test Outcomes

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Shared appendix referenced by MTDF Companion Papers Parts I to IV. Updated for Validation_ Dashboard_V74.html.

Abstract

This appendix consolidates the reproducibility index, test scripts, and summary diagnostics for the MTDF validation suite. It provides the complete inventory of tests referenced across the companion papers (MTDF_01 through MTDF_08), including script locations, input data, and pass/fail outcomes under the V18 strict protocol.

Claim status. This paper distinguishes between: (i) calibration anchors, (ii) validation targets, (iii) diagnostic consistency checks, and (iv) speculative extensions. Only results explicitly labelled as validation targets are used as evidence for the core MTDF claim. Diagnostic and speculative sections are included to define future tests, not to claim established confirmation.

1 Purpose and how to use this appendix

Data and code availability. The complete MTDF reproducibility package, including the modified CLASS solver (`class_mtdf`), the V18 strict validation workbook, the `run_validate.py` pipeline, the gravity-sector artefact manifest (`gravity/MANIFEST.sha256` with `VERIFY_MANIFEST.txt` witness), and the Phase 2 CosmoPower emulator cache, is archived on Zenodo (concept DOI 10.5281/zenodo.19741058; this release, `v1.1.2`, DOI 10.5281/zenodo.19743916) and mirrored on GitHub at github.com/IMesche/MTDF. The `v1.1.2` Git tag corresponds to the archived Zenodo release. Every script enumerated below is part of that archive and reproduces the outputs it references without further configuration, following the procedure in the top-level `README.md`.

This appendix consolidates the reproducibility index and quick-look outcomes for the test scripts used across the MTDF companion papers. The intent is simple: keep the core papers readable, while keeping the test inventory, scripts, and summary diagnostics in one stable place.

Files referenced here:

- Companion Part I: Environmental and local evidence
- Companion Part II: Gravity sector and lensing validation (*MTDF_03* [see companion paper])
- Companion Part III: Photon coupling, redshift, and early universe links
- Companion Part IV: Cosmological validation and high-energy test programme
- Interactive dashboard: Validation_Dashboard_V74.html

Scope note: This appendix is a reproducibility index. It does not change the evidential status of any result. Claim status is defined in the corresponding main or companion paper. Statistical weight remains with the distance-calibrated observables and the calibration versus validation separation defined in the dashboard workflow.

2 Test index and reproducibility

This appendix records the scripts used to generate the figures and summary diagnostics for Parts I to IV of the broader MTDF validation effort. The SN Ia × void result is treated as primary evidence in Companion Part I. The remaining tests are included as constraints, consistency checks, and forward-looking discriminants across Parts II and III.

2.1 Script index by paper component

Paper component	**Test**	**Script**	**Role**
Part I (environment and independent distances)	SN Ia × void analysis	sn_void_*.py	Primary evidence for an environment-dependent distance-calibrated residual.
	Zero-velocity shell	mtdf_zero_velocity_shell_test.py	Null and pipeline integrity check.
	Lensed image Δz	mtdf_lensing_redshift_split_test.py	Path-dependence test using multiple lensed images of the same source.
Part III (photon coupling κ)	*[speculative]*		
	Achromaticity	mtdf_achromaticity_test.py	Wavelength independence requirement for any stress-induced redshift.
	Dipole mismatch	mtdf_dipole_mismatch_test.py	Checks whether a stress-induced dipole can contribute to the quasar dipole anomaly.
	Dipole alignment	mtdf_dipole_vector_alignment.py	Directional consistency check between excess dipole and density gradient.
	Distance duality	mtdf_distance_duality_test.py	Etherington relation and potential environment split.
	Sandage-Loeb drift	mtdf_sandage_loeb_drift_test.py	Long-baseline, future definitive test of evolving redshift physics.
	Synthesis plot	mtdf_synthesis_plot.py	Conceptual summary figure linking multiple anomalies to the same scaling law.
	S8 tension and growth	mtdf_growth_S8_analysis.py	Large-scale structure growth consistency.
Part IV (cosmological and lensing)	BAO void split	mtdf_bao_void_split_test.py	Environment split for BAO-scale observables.
	GW standard sirens	mtdf_gw_siren_test.py	Independent distance ladder cross-check using gravitational waves.
	Lensing versus RSD (E_G)	mtdf_lensing_rsd_test.py	Modified-gravity consistency test, not a direct probe of photon coupling.
	Cluster mass MDR	mtdf_cluster_mass_discrepancy_test.py	Compares dynamical and lensing mass proxies in clusters.
	AP void shape	mtdf_ap_void_test.py	Alcock-Paczynski void geometry constraint.

Paper component	Test	Script	Role
Part II (gravity sector and lensing)	Strong lensing time delays	step20_strong_lensing.py	Time-delay distance consistency check under eta = 1.
	Elliptical galaxy dispersions	step22_elliptical_dispersions.py	Pressure-supported system check (SLACS-style).
	Satellite kinematics	step22b_satellite_kinematics.py	Digitised and table-based comparison (More+2011 style) with robustness notes.

2.2 Outcome snapshot

Script	Headline outcome	Interpretation at current sensitivity
mtdf_zero_velocity_shell_test.py	Consistent	No significant artifact detected in a designed null scenario.
mtdf_distance_duality_test.py	Consistent	No significant violation of $\eta = 1$ at current precision; environment split remains a future target.
mtdf_lensing_redshift_split_test.py	Consistent	No significant Δz between multiple lensed images; current spectroscopy is above the $\kappa \lesssim 10^{-3}$ signal level.
mtdf_achromaticity_test.py	Consistent at current precision	No evidence for chromatic redshift; any measured offsets are likely systematics at the present 10^{-5} level.
mtdf_dipole_mismatch_test.py	Suggestive (partial amplitude)	At $\kappa \approx 0.001$, MTDF reproduces the correct sign and roughly 42% of the reported quasar dipole excess amplitude.
mtdf_dipole_vector_alignment.py	Moderate directional agreement	Alignment is not tight enough to be a strong constraint, but it remains informative for future catalogues.
mtdf_sandage_loeb_drift_test.py	Future-only	Predicted environment split in $\dot{z}$ is far below current detection limits for the derived κ value ($f_{kick}/3$).
mtdf_synthesis_plot.py	Key synthesis figure	Heuristic summary connecting multiple anomalies across environments. It is not, by itself, a measurement of κ.
mtdf_lensing_rsd_test.py	Consistent	E_G data are broadly consistent with GR at current precision; not a direct κ probe.
mtdf_cluster_mass_discrepancy_test.py	Interesting signal (MDR $\approx$ 1.14)	Potential dynamical versus lensing mass tension; interpretation depends on mass calibration and baryonic systematics.
mtdf_ap_void_test.py	Consistent	Void shape residuals consistent with zero after RSD correction; κ effects are below current precision.
mtdf_growth_S8_analysis.py	See Part IV	Growth and S8 analyses are treated in the cosmological validation component.
mtdf_bao_void_split_test.py	See Part IV	BAO environment splits are treated as cosmological geometry constraints.
mtdf_gw_siren_test.py	See Part IV	GW sirens provide an independent distance check in the cosmological component.

Script	Headline outcome	Interpretation at current sensitivity
step20_strong_lensing.py	See Part II	Time-delay distance chiˆ2/nu = 0.77; isothermal slope gamma = 2.000 (structural prediction). See gravity-sector companion.
step22_elliptical_dispersions.py	See Part II	Mean velocity dispersion bias -1.4%; chiˆ2/nu = 1.81 (gas-corrected). See gravity-sector companion.
step22b_satellite_kinematics.py	See Part II	More+2011 amplitude test: mean |bias| 6.3%; profile chiˆ2/nu = 3.60 (best nuisance). See gravity-sector companion.

2.3 Synthesis figure

Figure A.1 provides a compact visual summary that places three anomalies on a single stress-scaling trend. This figure is intended as a high-level synthesis and motivates a unified search strategy. It should not be interpreted as a standalone parameter inference for κ.

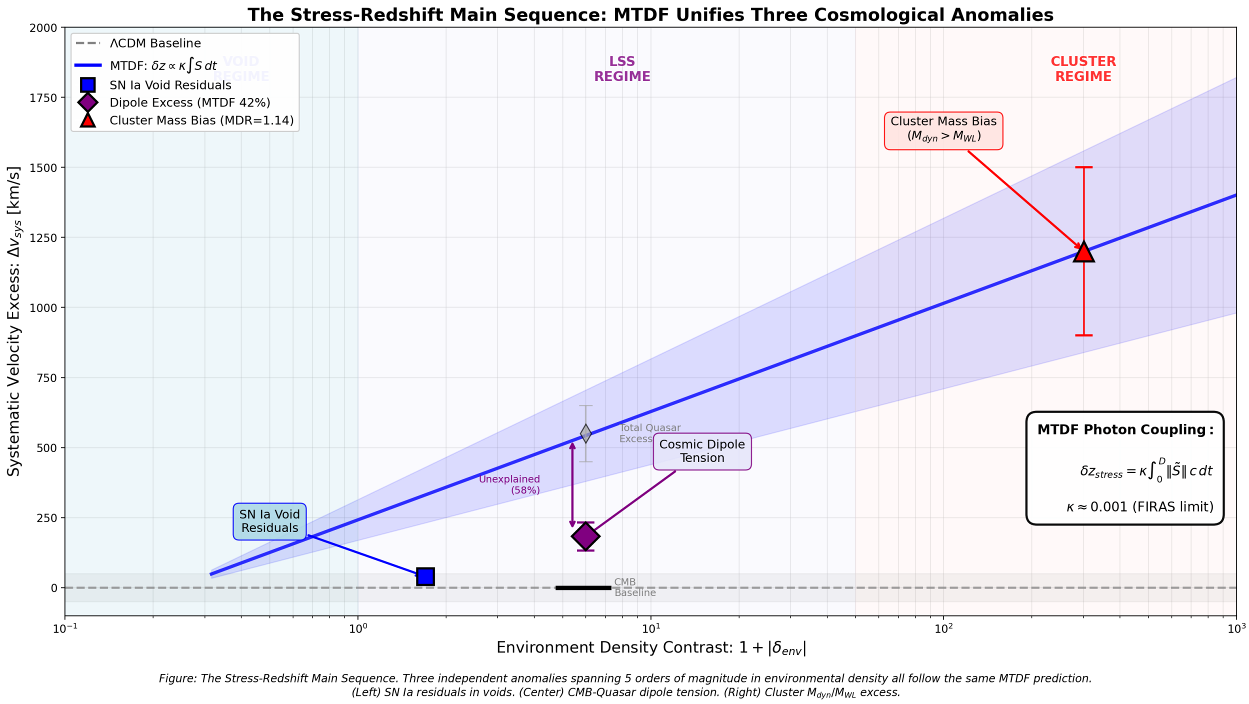

Figure A.1. The stress-redshift main sequence synthesis plot (generated by mtdf_synthesis_plot.py).